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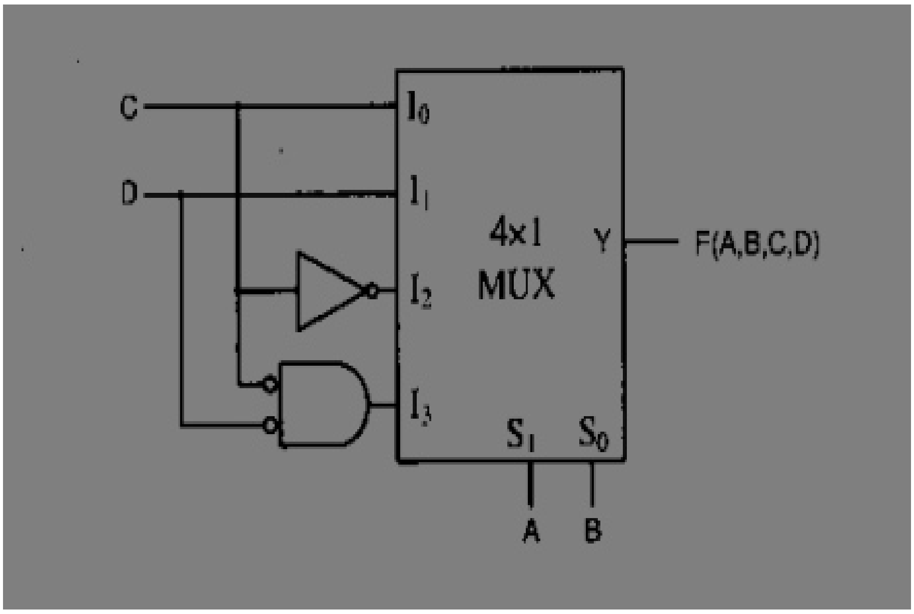
GATE 2010 EC, 39^{th} Question Analysis

Question 39

The Boolean function reaized by the logic circuit shown is



A $F = \sum m(0, 1, 3, 5, 9, 10, 14)$

B $F = \sum m(2, 3, 5, 7, 8, 9, 12, 13)$

C $F = \sum m(1, 2, 4, 5, 11, 14, 15)$

D $F = \sum m(2, 3, 5, 7, 8, 9, 12)$

Question Analysis:

- The standard mux equation of 4x1 mux is
- $F = \bar{S}_1.\bar{S}_0.I_0 + \bar{S}_1.S_0.I_1 + S_1.\bar{S}_0.I_2 + S_1.S_0.I_3$
- here $S_1 = A, S_0 = B, I_0 = C, I_1 = D, I_2 = \bar{C}, I_3 = \bar{C}.\bar{D}$.
- So $F = \bar{A}.\bar{B}.C + \bar{A}.B.D + A.\bar{B}.\bar{C} + A.B.\bar{C}.\bar{D}$
- The conanical form of F is as follows
- $F = \bar{A}.\bar{B}.C.\bar{D} + \bar{A}.\bar{B}.C.D + \bar{A}.B.\bar{C}.D + \bar{A}.B.C.D + A.\bar{B}.\bar{C}.\bar{D} + A.\bar{B}.\bar{C}.D + A.B.\bar{C}.\bar{D}$
- The Truth table for above expressions is as follows
- From Truth Table we can see that Option-D is correct.

The Truth Table

INPUTS				Output
A	B	C	D	$F(A,B,C,D)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here we used seven segment and 4 i/ps to display the corresponding output on seven segment display as per truth table and verified the expression.

Required Components & Pin Connections

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	Seven segment (1)
4	Resistors: 220Ω (1)
5	Jumper Wires
6	USB Cable

Component	Arduino Pin
Input A(GND or 5v)	Digital 5
Input B(GND or 5v)	Digital 4
Input C(GND or 5v)	Digital 3
Input D(GND or 5v)	Digital 2
Output a(segment a)	Digital 6
Output b(segment b)	Digital 7
Output c(segment c)	Digital 8
Output d(segment d)	Digital 9
Output e(segment e)	Digital 10
Output f(segment f)	Digital 11
Output g(segment g)	Digital 12
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $A = 0, B = 0, C = 0, D = 0$ by connecting them to ground
- $F = \bar{A}.\bar{B}.C + \bar{A}.B.D + A.\bar{B}.\bar{C} + A.B.\bar{C}.\bar{D}$

- No Change inputs A, B, C, D as per truth table by connecting to GND or 5V.
- Now Note Down the Experimental Truht Table

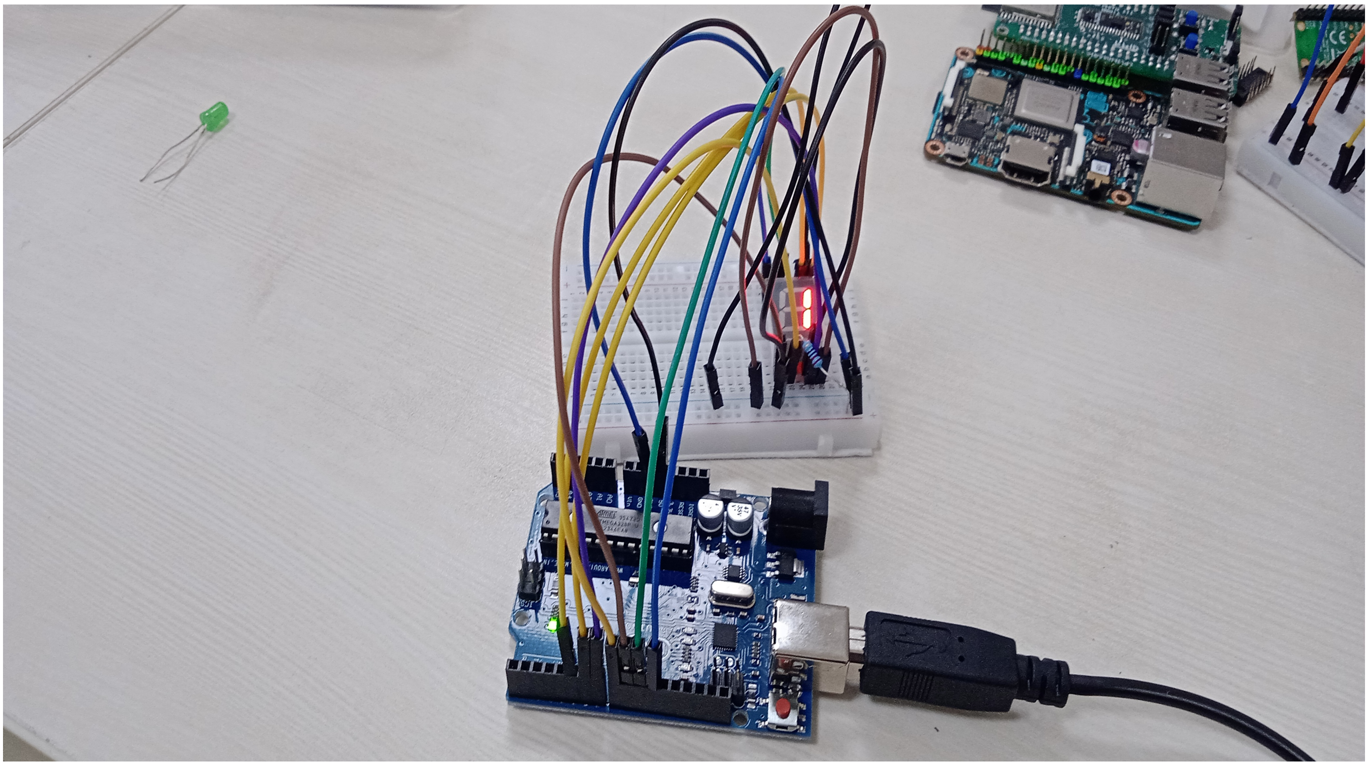
Code Uploading Steps

1. Create a new folder for BLINK
2. Write The code in main.asm
3. Run the main.asm with command "avra main.asm. It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precomplied" option
7. Observe the ouput and verify the expression

Experimental Truth Table

INPUTS				**Output**
A	B	C	D	$F(A, B, C, D)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Circuit



Conclusion

- By comparing Experimental Truth Table and Actual Truth table $F = \sum m(2, 3, 5, 7, 8, 9, 12)$.
- This matches option (D) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.